

```
In [1]: print("Hello, Introduction to Artificial Intelligence")
```

Hello, Introduction to Artificial Intelligence

```
In [2]: !pip install pandas
```

Collecting pandas

Downloading pandas-3.0.5-cp311-cp311-win_amd64.whl.metadata (19 kB)

Collecting numpy>=1.26.0 (from pandas)

Downloading numpy-2.4.6-cp311-cp311-win_amd64.whl.metadata (6.6 kB)

```
Requirement already satisfied: python-dateutil>=2.8.2 in F:\anaconda\install\envs\ai-lab\Lib\site-packages (from pandas) (2.9.0.post0)
```

Requirement already satisfied: tzdata in F:\anaconda install\envs\ai-lab\Lib\site-packages (from pandas) (2026.4)

Requirement already satisfied: six>=1.5 in F:\anaconda\install\envs\ai-lab\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

Downloading pandas-3.0.5-cp311-cp311-win_amd64.whl (10.0 MB)

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Downloading numpy-2.4.6-cp311-cp311-win_amd64.whl (12.6 MB)

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```

Installing collected packages: numpy, pandas

[illegible]

3/9

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[illegible]

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[illegible]

Successfully installed numpy-2.4.6 pandas-3.0.5

```
In [3]: import pandas as pd
print(pd.__version__)
```

3.0.5

```
In [4]: url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"
df = pd.read_csv(url)

print(df.shape)
```

 $(891, 15)$

```
In [5]: df.head()
```

Out[5]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T



In [6]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    str
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    str
8   class       891 non-null    str
9   who         891 non-null    str
10  adult_male  891 non-null    bool
11  deck        203 non-null    str
12  embark_town 889 non-null    str
13  alive       891 non-null    str
14  alone       891 non-null    bool
dtypes: bool(2), float64(2), int64(4), str(7)
memory usage: 92.4 KB
```

In [7]: `df.isnull().sum()`

```
Out[7]: survived      0
        pclass        0
        sex           0
        age          177
        sibsp         0
        parch         0
        fare          0
        embarked      2
        class         0
        who           0
        adult_male     0
        deck          688
        embark_town    2
        alive          0
        alone          0
        dtype: int64
```

```
In [8]: df.describe()
```

```
Out[8]:
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [9]: numeric = df.select_dtypes(include="number")
        cv = (numeric.std() / numeric.mean()).abs()
        print(cv)
        print("Largest:", cv.idxmax(), cv.max())
```

```
survived      1.267701
pclass         0.362149
age           0.489122
sibsp         2.108464
parch         2.112344
fare          1.543073
dtype: float64
Largest: parch 2.1123440706670116
```

```
In [ ]:
```